



Experimental and kinetic modeling study of 2-methyltetrahydrofuran oxidation under engine-relevant conditions



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ABSTRACT

The oxidation of 2-methyltetrahydrofuran (2-MTHF) has been experimentally studied in a rapid compression machine. The ignition delays of stoichiometric 2-MTHF/O₂/inert mixtures were measured for a wide range of conditions: Top dead center pressure ranging from 0.3 to 2.1 MPa and core gas temperatures between 640 K and 900 K. Two-stage ignition is observed for temperatures between 685 K and 790 K, leading to a deviation from Arrhenius behavior of the evolution of ignition delays with temperature. Sampling of the reacting mixture was performed for different times between the top dead center and the ignition. Stable reaction intermediates were separated by gas chromatography, identified by mass spectrometry and quantified with a flame ionization detector. More than 20 species were identified and used to identify major reaction pathways. Among species of importance, 2-methyl-dihydrofurans, 2-methylfuran, methyl-dihydrofuranones, 1-cyclopropanylethanone, 3-penten-2-one and methylvinylketone were detected. Mole fraction profiles were obtained and used alongside first stage and total ignition delays to validate a detailed kinetic model for the oxidation of 2-MTHF. The resulting model takes in account 507 species involved in 2425 reactions. The validity of this model was extended to the high temperature range using existing ignition delays and low pressure laminar flame speciation data.

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1. Introduction

The production of liquid fuels from lignocellulosic biomass is currently considered a potential source of biofuels that are non-dependent on edible resources. Recent advances in catalytic conversion of biomass show high yield options for the production of species like levulinic acid [1]. This compound can further be converted into a variety of species, including 2-methyltetrahydrofuran (2-MTHF) [2]. The use of 2-MTHF as a fuel has therefore been considered in several studies, for both SI and CI engine technologies [3–5].

2-MTHF has a relatively low Research Octane Number of 86, which may limit its application as a blendstock in modern SI engines [3], but an energy density of 34.1 kJ/kg, which is higher than ethanol (26.8 kJ/kg), and a boiling point of 78–80 °C. The feasibility of blending it up to 40% in gasoline has been investigated in the literature [4], the higher blending ratios leading to a significant reduction in hydrocarbons and benzene emissions, but an increase in NO_x emissions. The potential of using a 2-MTHF/dibutylether mixture as a synthetic diesel biofuel has also been investigated

in a single cylinder engine [5]. The authors noticed that with this fuel blend, particle emissions could be significantly reduced even at the greatest EGR rates.

Since 2010, several studies have been published on the oxidation of 2-MTHF. Simmie [6] calculated the main kinetic rate constants for the first steps of the oxidation of 2,5-dimethyltetrahydrofuran (DMTHF) and 2-MTHF at high temperatures. Bond dissociation energies (BDE) for 2-MTHF were calculated and are summarized in Fig. 1.

Moshhammer et al. [7] studied the combustion of 2-MTHF in low pressure flat flames. Mole fraction profiles of the intermediate species and radicals formed in the flame are provided. Analyses were performed by Molecular Beam Mass Spectrometry. The authors also proposed a model of the oxidation of 2-MTHF validated in low pressure flame conditions, including the rate constants from Simmie's theoretical study. Wang et al. [8] investigated the ignition delays of 2-MTHF in a shock tube under different conditions. The authors compared experimental results and simulations obtained using the model published by Moshhammer et al., showing a good agreement. Chakravarty et al. [9] published a theoretical study of the reaction kinetics of hydrogen abstraction reactions by hydroperoxyl radical from 2-MTHF and DMTHF, which are critical reactions for the modeling of ignition delays in the low to intermediate temperature regime. Sudholt et al. [10] calculated

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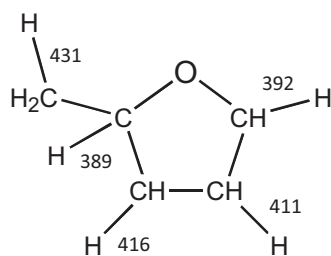


Fig. 1. C–H bond dissociation energies calculated by Simmie [6] for 2-MTHF, in kJ mol^{-1} .

the BDE for a range of saturated and unsaturated (hydro) furans, among which 2-MTHF. Their results are comparable to those reported by Simmie [6].

To our knowledge, no experimental study addresses the low-temperature oxidation of 2-MTHF. This study therefore aims to provide reliable experimental data on 2-MTHF ignition, in the low-temperature regime and at high pressures. To do so, first stage and total ignition delays and mole fraction profiles of stable intermediates of stoichiometric 2-MTHF/ O_2 /inert mixtures were measured in the U. Lille Rapid Compression Machine (RCM), at compressed pressures and temperatures ranging from 0.3 to 2.1 MPa and 640 to 900 K, respectively. These data served to validate a kinetic model, which was also validated in the high temperature regime by simulating experimental data found in the literature. This follows an experimental study of tetrahydrofuran (THF) performed on the same RCM [11], in an attempt to provide insight on the reaction pathways associated with the two-stage ignition of a series of cyclic ethers.

2. Experimental methods

The U. Lille RCM has been described in detail previously [12–14], so only the most important elements are summarized in the following section. This RCM has a right angle design, allowing to keep the combustion chamber volume strictly constant at top dead center and to control the piston velocity throughout the whole compression phase. For this study, the volumetric compression ratio was set to 9.65. The compression time was chosen to be 60 ms to minimize the onset of turbulence after compression, following previous work [14]. The reaction chamber is covered by heating elements ensuring a homogeneous initial temperature with a deviation of ± 1 K along the axial dimension.

Pressure profiles are recorded by a Kistler 6052 piezoelectric pressure transducer coupled with a 5007 charge amplifier with 40 μs time step. The ignition delay time is defined as the time interval between the end of the compression (TDC: Top Dead Center) and the maximum dp/dt associated with ignition (see Fig. 2). The first stage ignition delay time is defined as the time interval between TDC and the maximum dp/dt in the pressure jump which corresponds to the first stage ignition. This time has been seen to correspond to the maximal light emission associated with the first stage ignition in other two stage ignition experiments [15].

The core gas temperature is calculated following the adiabatic core assumption, from the pressure at top dead center and the initial conditions [16]. A creviced piston head is used, following the recommendations of Lee and Hochgreb [17], to ensure maximal homogeneity of the temperature at the end of the compression. The pressure oscillations near TDC are due to slight vibrations of the combustion chamber after the compression event. They result in an estimated uncertainty of ± 5 K on the calculated core gas temperature.

The reactant mixtures were prepared via the partial pressures method and left to homogenize overnight. All the mixtures were

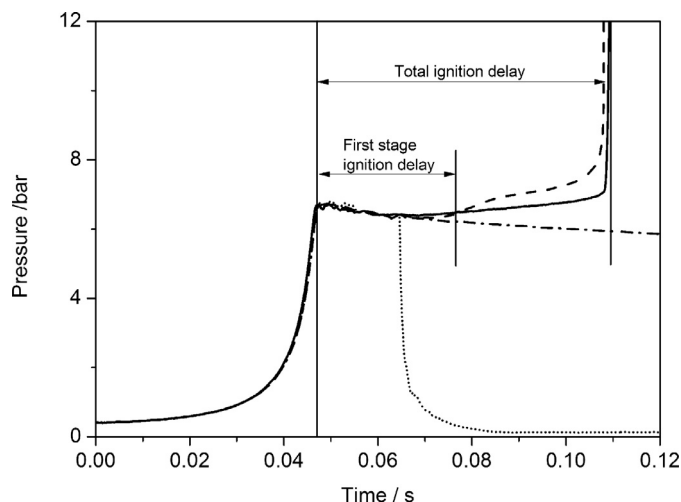


Fig. 2. Pressure profiles for 2-MTHF/ O_2 / N_2 stoichiometric mixtures, $p_{\text{TDC}} = 0.67$ MPa and $T_{\text{C}} = 719$ K. Dotted-dashed line: unreactive mixture; continuous line: reactive mixture; dashed line: simulation using the current model; Dotted line: example of pressure profile during a sampling experiment.

stoichiometric. Anhydrous stabilizer-free 2-MTHF (purity > 99%) was acquired from Sigma-Aldrich. Purity of oxygen O_2 , N_2 , Ar and CO_2 , provided by Air Liquide, is 99.99%. Inert gas composition was adjusted in order to vary the core gas temperature, due to differences in heat capacities for the N_2 , Ar and CO_2 . The reactant mixture compositions and the obtained values of T_{C} are listed in Table 1. A detailed table summarizing all the ignition delay experiments is provided in the supplementary material.

Nonreactive pressure profiles were recorded for each mixture by replacing O_2 by N_2 , in order to characterize the pressure decrease due to the boundary layer expansion after the end of compression. These nonreactive profiles are used for zero-dimensional simulations, as described in computational methods section, and are available in the Senkin input file format in the Supplementary material. The experiments were repeated a minimum of 3 times. The overall reproducibility of the measurements was very good, with first stage and total ignition delays being reproducible within the range of 2 ms, except for the longest ignition delays (i.e. superior to 100 ms).

Sampling of the reactive mixture was performed by puncturing a diaphragm at selected times during the ignition delay. The mixture was then depressurized into a heated canister. The pressure dropped to half its initial value in less than a millisecond, offering instantaneous cooling of the gaseous mixture and good temporal resolution. An example of the pressure profile during a sampling experiment is shown in Fig. 2. Then, the canister contents were slowly compressed, for better sensitivity, and injected in a gas chromatograph (Bruker – Scion SQ 456) equipped with dual TCD/FID and MS detection. The chromatograph is equipped with HP5 and Porabond Q columns. Porabond Q was estimated more specific for species separation in those experiments. Quantification of stable reaction intermediates was performed with the FID. The FID responses of 2-MTHF, Oxetane, butanol, acetone and 2-MTHF were duly calibrated. For the other quantified compounds, calibration factors were deduced from the effective carbon number concept [18]. Uncertainty estimates on obtained mole fractions were about $\pm 5\%$ for species calibrated using standards and $\pm 10\%$ for species calibrated using the effective carbon number method, as is usually reported for gas chromatography techniques [11]. The reproducibility of the sampling technique is visible from the mole fraction profiles displayed in Fig. 8. The carbon balance obtained before the first stage delay was superior to 90%. After the first

Table 1
Mixture compositions and associated mean core gas temperature.

Mole fraction 2-MTHF	Mole fraction O ₂	Mole fraction CO ₂	Mole fraction Ar	Mole fraction N ₂	Mean T _c /K
0.02913	0.20388	0.41417	–	0.35282	644
0.02913	0.20388	0.28379	–	0.48320	664
0.02913	0.20388	0.17641	–	0.59058	685
0.02913	0.20388	0.07670	–	0.69029	707
0.02913	0.20388	–	–	0.76699	719
0.02913	0.20388	–	0.15340	0.61359	749
0.02913	0.20388	–	0.24544	0.52155	761
0.02913	0.20388	–	0.38350	0.38350	790
0.02913	0.20388	–	0.49087	0.27612	815
0.02913	0.20388	–	0.57524	0.19175	835
0.02913	0.20388	–	0.65194	0.11505	847
0.02913	0.20388	–	0.71330	0.05369	865
0.02913	0.20388	–	0.76699	–	886

stage ignition, it was reduced to about 60%, the quantification of CO, CH₂O and CO₂ being impossible with our experimental method.

3. Computational methods

A hierarchical detailed kinetic model was built in order to describe the oxidation of 2-MTHF in both low and high temperature regions. Table 2 shows the nomination of some representative species in the 2-MTHF sub mechanism.

The C₀–C₄ base of the mechanism used in this work is derived from the work of Metcalfe et al. [19], which has been validated on RCM and shock tube data on numerous occasions.

The high temperature sub-mechanism for the oxidation of 2-MTHF published by Moshhammer et al. [7] contains the recent set of reaction rates calculated by Simmie [6], including H-abstractions from 2-MTHF by H[•] radical and CH₃[•] radical, H-elimination from 2-methyltetrahydrofuran yielding corresponding 2-methyldihydrofurans and ring opening of 2-methyltetrahydrofuran by beta-scission. Moshhammer et al.'s [7] sub-mechanism is included in the current mechanism with some updates detailed below. First, the recent values for the reaction rate constants reported by Chakravarty et al. [9] for H-abstraction reaction by hydroperoxyl radical yielding 2-methyltetrahydrofuran radicals (MTHFyl[•]) and hydrogen peroxide, 2-MTHF + HO₂[•] ⇌ MTHFyl[•] + H₂O₂, were used. The rate constant parameters for H-abstraction reactions are summarized in Table 3.

Also, bimolecular reactions 2-MTHF + O₂ ⇌ MTHFyl[•] + HO₂[•] were included, using rate constants suggested by Pitz et al. [21] from their methylcyclohexane study. A reduction of the activation energy of 4 kcal/mol was made in the case of the abstraction of H atoms carried by carbon atoms bonded to the oxygen atom (Table 4).

Finally, three reaction rates for the H-abstraction from the fuel by [•]OH radicals proposed by Moshhammer et al. [7] were modified. These reactions lead to 5 different 2-methyltetrahydrofuran radicals and are among the most sensitive reactions on the first stage and total ignition delays, as depicted in Fig. 6. In Moshhammer et al.'s mechanism, these reaction rates are equal for the formation of CH₂MTHF, R2MTHF, R3MTHF and R4MTHF radicals, which can lead to an inaccurate branching ratio. The reaction rates for the formation of R3MTHF and R4MTHF from Moshhammer et al. and Pitz et al. [21] were conserved. The rate constant of the H-abstraction reaction that forms CH₂MTHF was taken from the work of Pitz et al. [21] for H abstraction by [•]OH on the methyl group of methylcyclohexane. Then, for the H-abstraction that leads to R2MTHF, where the carbon atom is bonded to the oxygen atom, the reaction rate of Pitz et al. for the tertiary position of methylcyclohexane was included, with a reduction of the activation energy by 345 cal/mol to take into account the influence of the oxygen

Table 2
Nomination of some representative species in the 2-MTHF sub mechanism.

MTHF-2	
CH ₂ MTHF	
R2MTHF	
R3MTHF	
R4MTHF	
R5MTHF	
O ₂ CCYCCCCO	
CCYC(O ₂)CCCO	
CCYCC(O ₂)CCO	
CCYCCC(O ₂)CO	
CCYCCCC(O ₂)O	
CYC [•] CCC(C)O	
CJCCC([•] O)C	
CCYC4OKET35	
CCYCJCC(O ₂ H)CO	
CCYCCO ₂ CCO ₂ HO	

Table 3

H-abstraction reactions rate constant parameters used in the present study. The Arrhenius parameters are given in units of s^{-1} , cm^3 , mol^{-1} , cal.

Reaction	A	n	E_a	Ref
MTHF-2 + H=CH ₂ MTHF+H ₂	1.07E + 03	3.064	6.571	[6] ^a
MTHF-2 + H=R2MTHF+H ₂	1.96E + 05	2.354	637.7	[6] ^a
MTHF-2 + H=R3MTHF+H ₂	1.69E + 03	3.005	3.906	[6] ^a
MTHF-2 + H=R4MTHF+H ₂	1.69E + 03	3.005	3.906	[6] ^a
MTHF-2 + H=R5MTHF+H ₂	5.80E + 05	2.49	2330	[6] ^a
MTHF-2 + CH ₃ =CH ₂ MTHF+CH ₄	4.87E + 02	2.886	2.886	[6] ^a
MTHF-2 + CH ₃ =R2MTHF+CH ₄	1.22E + 03	2.818	2.818	[6] ^a
MTHF-2 + CH ₃ =R3MTHF+CH ₄	0.54E + 03	2.887	2.887	[6] ^a
MTHF-2 + CH ₃ =R4MTHF+CH ₄	0.54E + 03	2.887	2.887	[6] ^a
MTHF-2 + CH ₃ =R5MTHF+CH ₄	1.22E + 03	2.818	2.818	[6] ^a
MTHF-2 + O=CH ₂ MTHF+OH	3.550E + 06	2.400	5830	[20] ^b
MTHF-2 + O=R2MTHF+OH	5.620E + 13	0	5200	[20] ^b
MTHF-2 + O=R3MTHF+OH	1.3E + 12	0	5200	[27] ^c
MTHF-2 + O=R4MTHF+OH	1.3E + 12	0	5200	[27] ^c
MTHF-2 + O=R5MTHF+OH	5.620E + 13	0	5200	[20] ^b
MTHF-2 + OH=R2MTHF+H ₂ O	5.733E + 10	0.51	−280	[21] ^d
MTHF-2 + OH=R3MTHF+H ₂ O	4.670E + 07	1.61	−35	[21] ^e
MTHF-2 + OH=R4MTHF+H ₂ O	4.670E + 07	1.61	−35	[21] ^e
MTHF-2 + OH=R5MTHF+H ₂ O	4.670E + 07	1.61	−380	[21] ^d
MTHF-2 + OH=CH ₂ MTHF+H ₂ O	5.268E + 09	0.97	1586	[21]
MTHF-2 + HO ₂ =CH ₂ MTHF+H ₂ O ₂	1.66E + 02	3.47	17,080	[9]
MTHF-2 + HO ₂ =R2MTHF+H ₂ O ₂	6.52E + 01	3.27	6410	[9]
MTHF-2 + HO ₂ =R3MTHF+H ₂ O ₂	1.18E + 00	3.72	12,160	[9]
MTHF-2 + HO ₂ =R4MTHF+H ₂ O ₂	7.21E−02	4.13	11,460	[9]
MTHF-2 + HO ₂ =R5MTHF+H ₂ O ₂	7.79E + 00	3.53	7890	[9]

^a From (2,5)-dimethyltetrahydrofuran.

^b From diethylether.

^c From cyclopentane.

^d From methylcyclohexane −345 cal/mol to E_a to take into account O in beta position from abstracted H atom.

^e From methylcyclohexane.

Table 4

Bimolecular reactions between 2-MTHF and O₂ rate constant parameters used in the present study. The Arrhenius parameters are given in units of s^{-1} , cm^3 , mol^{-1} , cal.

Reaction	A	n	E_a	Ref
MTHF-2 + O ₂ =R2MTHF+HO ₂	1.000E + 13	0	44,200	[21] ^a
MTHF-2 + O ₂ =R3MTHF+HO ₂	2.000E + 13	0	49,640	[21] ^b
MTHF-2 + O ₂ =R4MTHF+HO ₂	2.000E + 13	0	49,640	[21] ^b
MTHF-2 + O ₂ =R5MTHF+HO ₂	2.000E + 13	0	45,640	[21] ^a
MTHF-2 + O ₂ =CH ₂ MTHF+HO ₂	3.000E + 13	0	52,290	[21]

^a From methylcyclohexane −4 kcal/mol to E_a to take into account O in beta position from abstracted H atom.

^b From methylcyclohexane.

atom. The rate constant for the formation of R5MTHF was adapted from the rate constant we described for the formation of R3MTHF and R4MTHF, the activation energy being decreased by 345 cal/mol in order to take into account the influence of the O atom. With this rate constant data, at a temperature of 700 K, the reaction rate of 2-MTHF + $\cdot\text{OH} \rightleftharpoons \text{R5MTHF} + \text{H}_2\text{O}$ is slightly superior to the reaction rate of 2-MTHF + $\cdot\text{OH} \rightleftharpoons \text{R2MTHF} + \text{H}_2\text{O}$, itself superior to the reaction rate of 2-MTHF + $\cdot\text{OH} \rightleftharpoons \text{R3MTHF/R4MTHF} + \text{H}_2\text{O}$, which is superior by a factor of 2 at 700 K to the reaction rate of 2-MTHF + $\cdot\text{OH} \rightleftharpoons \text{CH}_2\text{MTHF} + \text{H}_2\text{O}$. This matches with the expected branching ratio from the bond dissociation energies displayed in Fig. 1.

Due to the sparsity of kinetic modeling studies of 2-MTHF in the low temperature regime, there is a lack of rate constant data for the associated reaction pathways. Following the well described kinetic scheme of the low temperature oxidation of alkanes [22], we considered reaction pathways of 2-MTHF in this temperature range and estimated reaction rate constants. After their formation from H-abstraction reactions on 2-MTHF by O₂, OH $\cdot$ or HO₂ $\cdot$, MTHFyl $\cdot$ radicals can add to molecular oxygen (Tables 3 and 5).

Table 5

Reactions of addition of a methyltetrahydrofuranyl radical on O₂ rate constant parameters used in the present study. The Arrhenius parameters are given in units of s^{-1} , cm^3 , mol^{-1} , cal.

Reaction	A	n	E_a	Ref
O ₂ CCYCCCCO=CH ₂ MTHF+O ₂	5.93E + 22	−2.60	36,500	[23]
CCYC(O ₂)CCCO=R2MTHF+O ₂	1.54E + 28	−3.98	40,200	[23]
CCYCC(O ₂)CCO=R3MTHF+O ₂	9.74E + 26	−3.62	38,600	[23]
CCYCCC(O ₂)CO=R4MTHF+O ₂	4.30E + 26	−3.55	38,600	[23]
CCYCCCC(O ₂)O=R5MTHF+O ₂	9.74E + 26	−3.62	38,600	[23]

Two reaction pathways are in competition: formation of the corresponding unsaturated cyclic ether (methylidihydrofuran or methylenetetrahydrofuran) and HO₂ $\cdot$ radical, or of a peroxy-methyltetrahydrofuranyl radical MTHFylOO $\cdot$, the latter being preferred in the temperature range 650–750 K. For both reaction pathways, reaction rates calculated by Villano et al. [23] were used. Peroxy-methyltetrahydrofuranyl radicals MTHFylOO $\cdot$ can then react through two channels: First, by H transfer from HO₂ $\cdot$ radicals to MTHFylOO $\cdot$ (MTHFylOO $\cdot$ + HO₂ $\cdot$ $\rightleftharpoons$ MTHFylOOH + O₂) as described by Curran et al. [24]. MTHFylOOH then undergoes an O–O bond breaking to form MTHFylO $\cdot$ and OH $\cdot$ radical. Reaction rates for those reactions, as well as for the decomposition of MTHFylO $\cdot$ were adapted from corresponding reactions in the methylcyclohexane mechanism [21]. The second channel, which is very important between 650 and 750 K, is the isomerization of MTHFylOO $\cdot$, giving hydroperoxymethyltetrahydrofuranyl ($\cdot\text{QO}_2\text{H}$) radicals.

These radicals can yield methylidihydrofurans or 2-methylenetetrahydrofuran, methyltetrahydrofuranones or dicyclic diethers. The formation of methyltetrahydrofuranone isomers is described using the reaction rates calculated by Sirjean et al. [25] for reactions of cyclopentanyl radical + O₂ giving cyclopentanone + OH $\cdot$. Dicyclic diethers are likely to either isomerize into cyclic carbonyls, or to react through H abstraction by OH $\cdot$ or HO₂ $\cdot$ and readily decompose into linear radicals, finally decomposing through β -scission reactions, as described in the mechanism published by Sarathy et al. [26].

Nevertheless, the addition of hydroperoxymethyltetrahydrofuranyl radicals to O₂ (second O₂ addition) is favored. The formed $\cdot\text{O}_2\text{QO}_2\text{H}$ radicals lead to the low temperature chain branching through the formation of two $\cdot\text{OH}$ radicals. These reactions are included in the mechanism with reaction rate constants adapted from those published by Pitz et al. for methylcyclohexane [21].

In the intermediate temperature range (i.e. $T > 800$ K), the production of methylidihydrofurans from MTHFyl $\cdot$ + O₂ becomes the fastest consumption path of MTHFyl $\cdot$ radicals, and the reaction pathways of these species were included in the mechanism, using an analogy with the oxidation of cyclopentene described by Sirjean et al. [27], to yield 2-methylfuran (2-MF). In order to take into account the oxidation of 2-MF, the submechanism of Somers et al. [28] was added. Finally, the rate constants suggested by Lifshitz et al. [29] for the isomerization of 5-methyl-(2,3)-dihydrofuran into cyclopropylethanone and pent-4-en-2-one were included in the kinetic model.

The premixed low pressure flat flames simulations were computed with the subroutine PREMIX [30]. Note that only the high temperature section of the model was kept for these simulations to enable convergence in the computation, with an adaptive-grid-control-based solution curvature and gradient of 0.1. Removed classes of reactions in the high temperature mechanism are the addition of 2-methyltetrahydrofuranyl radicals to O₂ leading to the formation of hydroperoxyl radicals, and all further reactions of the hydroperoxyl radicals formed by the previous sequence. The RCM and shock tube results were simulated with the subroutine SENKIN [31] of the CHEMKIN package [32]. In order to take into account heat losses and the compression phase during the RCM

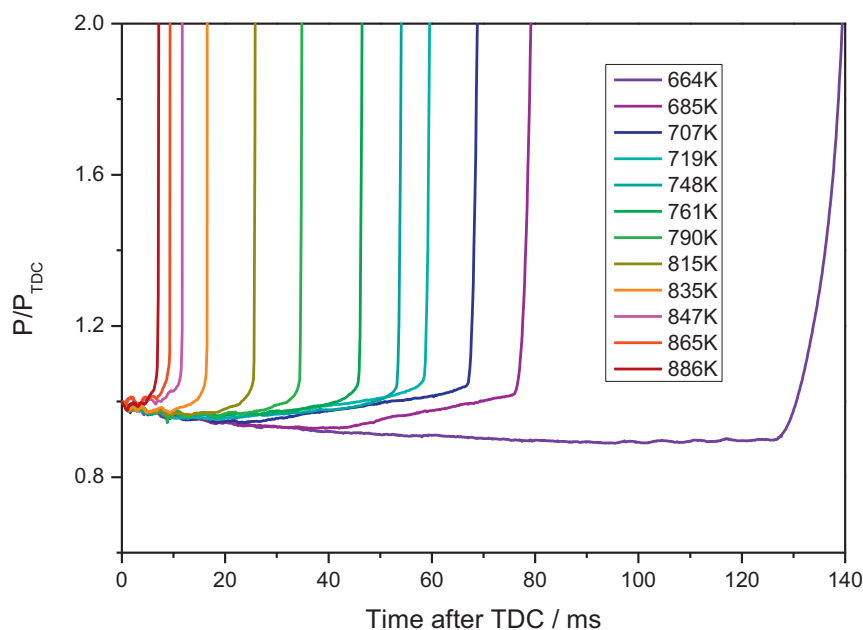


Fig. 3. Normalized pressure profiles for stoichiometric 2-MTHF/O₂/inert mixtures of ignition at different core as temperatures T_c . $p_{TDC} = 0.63$ – 0.80 MPa.

experiments, nonreactive profiles were used in order to extract volume profiles for the adiabatic core using the isentropic law [33]. These volume profiles are input files for calculations, and are provided in the Supplementary material. The thermochemical properties of all the new species included in the low temperature mechanism were evaluated using the THERM program developed by Ritter and Bozzelli [34], using the original group values.

The resulting detailed kinetic mechanism consists of 507 species involved in 2425 reactions, and enables simulation of the oxidation of 2-MTHF in a wide range of pressures and temperatures. It is included in the supplementary material in the CHEMKIN format.

4. Results and discussion

4.1. Ignition delays

In this study, total and first stage (when applicable) ignition delays of stoichiometric 2-MTHF/O₂/inert mixtures were measured at core gas temperatures and compressed pressures between 640 and 900 K, and 0.3 and 2.1 MPa, respectively. Two-stage ignition was observed for core gas temperatures ranging from 685 to 790 K. Figure 3 displays unfiltered normalized pressure profiles showing transition from two-stage to single-stage ignition at temperatures superior to approximately 820 K. The first stage ignition is faint, consistent with our previous study of THF [11], and is associated with limited low temperature heat release. First stage ignition delays are therefore reported only when their measurement was possible. In the case of alkanes, first stage ignition usually results in stronger heat release, leading to a more visible pressure jump. We attribute this difference in phenomenology to a combination of mass transfer to the crevice [35] and kinetic effects specific of cyclic ethers. Pre-ignition events can be ruled out as a consequence of the high reproducibility of the data and the quick consumption of the fuel and formation of intermediates during first stage ignition. This differs from mole fraction profiles observed for propagative cool flames in the same RCM [36].

Figure 2 shows a comparison with the experimental and modeled pressure profiles for two-stage ignition at p_{TDC} of 0.67 MPa and T_c of 719 K. One can see that both the first stage and total

ignition delays are fairly well reproduced, leading to good agreement between the experimental and modeled pressure profiles. The pressure increase associated with the first stage ignition is slightly overestimated. This is very likely due to limited mass transfer to the piston crevice, as described in [35]. Figure 4 re-groups the entire experimental data for ignition and first stage delays. Only delays between 2 and 200 ms, the typical accepted measurement domain for our RCM, are reported.

Globally, ignition delay times decrease with pressure and temperature. First stage ignition can be observed for temperatures between 685 and 790 K. The overall reproducibility of the measurement is good, but the experimental scatter increases for long delays.

The accurate prediction of two-stage ignition delays requires that the model can reproduce both the first stage and total ignition delay. A kinetic model dedicated to the simulation of such experiments should therefore be validated on both these targets. The present kinetic model was therefore used to predict the first stage and total ignition delay data measured in this work. Figure 5 shows a comparison between the evolution of experimental and modeled ignition delay times as a function of the core gas temperature, at four different initial pressures covering compressed pressures p_{TDC} ranging from 6.3 to 1.57 MPa. In each case, the model is able to simulate first stage ignition delays with reasonable agreement, the biggest deviations being observed for the lowest temperatures. When plotted against temperature, ignition delays exhibit a deviation from Arrhenius behavior between 750 and 850 K, this trend being well captured by the model at all pressures. The agreement between experimental and simulated ignition delay is fair, except for the lowest temperatures and highest pressures, with the highest discrepancy being of a factor of 2.

In order to highlight the important reactions involved in 2-MTHF oxidation, a “brute force” sensitivity analysis on both first stage and total ignition delays was performed. The sensitivity coefficient on total ignition delay of reaction i (σ_i) is defined as: $\sigma_i = \log(\tau_i^+/\tau_i^-) / (\log(2.0/0.5))$, where τ_i^+ is the total ignition delay calculated with a factor of two increase in the reaction rate of reaction i (k_i), and τ_i^- is the ignition delay calculated with a factor 2 decrease in k_i . The sensitivity coefficient on first stage ignition delay is calculated with the same process, changing total

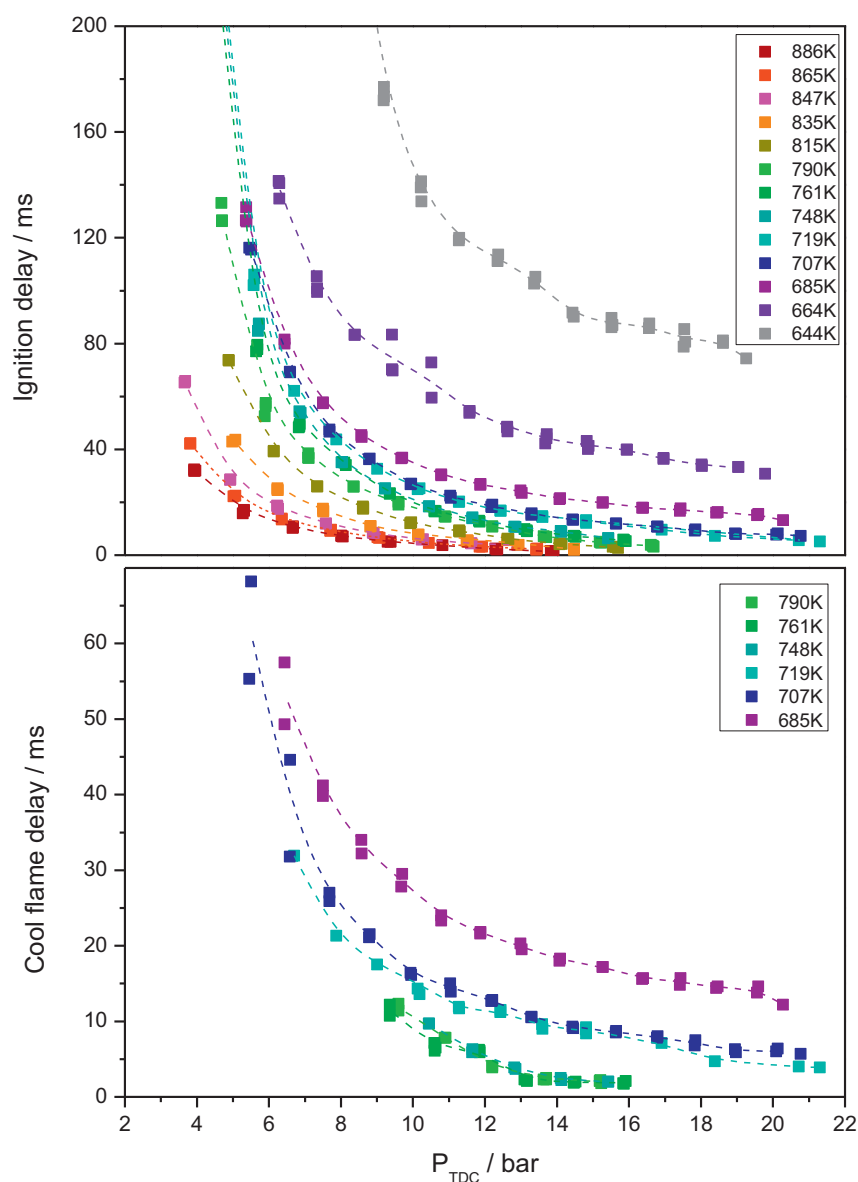


Fig. 4. Experimental first stage and total ignition delays of stoichiometric 2-MTHF/O₂/inert mixtures as a function of the top dead center pressure, for different core gas temperature T_c . Dashed line is a β -spline fit of the data.

ignition delays by first stage ignition delays. A positive σ_i indicates an overall inhibiting effect on reactivity, and vice versa.

Figure 6 shows that the system is sensitive to the branching ratio of formed methyltetrahydrofuranyl radicals. The reaction $\text{MTHF-2} + \cdot\text{OH} \rightleftharpoons \text{R2MTHF} + \text{H}_2\text{O}$ has a great positive sensitivity coefficient. In contrast, the sensitivity of $\text{MTHF-2} + \cdot\text{OH} \rightleftharpoons \text{R3MTHF} + \text{H}_2\text{O}$ is negative. It is due to the impossibility of R2MTHF to form a ketohydroperoxyde after second O₂ addition. This radical mainly undergoes an addition on O₂ giving a ROO $\cdot$ radical that decomposes into HO₂ $\cdot$ and 5-methyl-(2,3)-dihydrofuran (Fig. 9), an unsaturated compound with lower reactivity than 2-MTHF. In contrast, R3MTHF reacts through the low-temperature indirect chain branching pathway. Reactions of 2-MTHF with $\cdot\text{OH}$ yielding R2MTHF, R4MTHF or R5MTHF and H₂O exhibit a negative sensitivity coefficient but do not appear in Fig. 6 due to lower sensitivity coefficients. Also, reactions of ROO $\cdot$ leading to the formation of methylidihydrofurans and hydroperoxyl radical show a positive sensitivity. In the other hand, the reactions of ROO $\cdot$

isomerizing into $\cdot\text{QOOH}$ show a negative sensitivity. The latter leads to the low-temperature branching pathway, whereas the former leads to chain termination, causing an inhibition of the overall reactivity. The ignition delay is finally sensitive to the isomerization of $\cdot\text{O}_2\text{QO}_2\text{H}$ radicals. This indicates that these radicals undergo typical low-temperature chain branching reactions, except for $\cdot\text{O}_2\text{QO}_2\text{H}$ radical formed via R2MTHF radicals.

It is interesting to note that the sensitivity analysis on the total ignition delay reveals a similar trend to the sensitivity analysis on the first stage ignition delay. This demonstrates that in these temperature and pressure conditions, accurate prediction of the low temperature kinetics and first stage ignition is critical if agreement is sought on the total ignition delay. The reaction $\text{HO}_2\cdot + \text{HO}_2\cdot \rightleftharpoons \text{H}_2\text{O}_2 + \text{O}_2$ shows a great positive coefficient due to the consumption of two HO₂ $\cdot$ radicals. However, the reaction of HO₂ $\cdot$ with 2-MTHF leading to the formation of R2MTHF has an important promoting effect on the reactivity of the system, as it is one of the main sources of H₂O₂ and only consumes one HO₂ $\cdot$ radical.

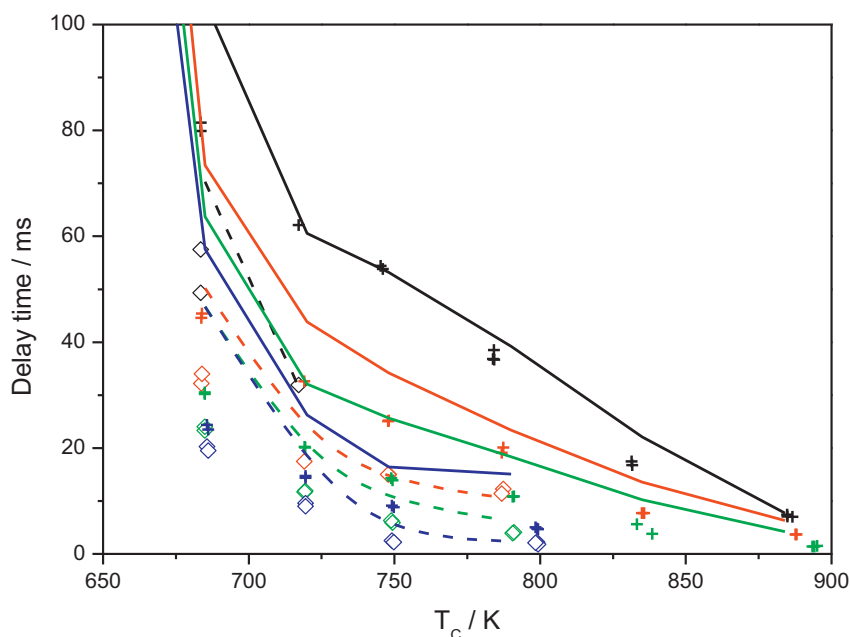


Fig. 5. Comparison of the experimental and simulated first stage ignition (open symbols, dashed line) and total ignition delays (full symbols, continuous line) of stoichiometric 2-MTHF/O₂/inert mixtures for (black) $p_0 = 40.0$ kPa, $p_{TDC} = 0.63$ – 0.8 MPa; (red) $p_0 = 53.3$ kPa, $p_{TDC} = 0.82$ – 1.08 MPa; (green) $p_0 = 66.7$ kPa, $p_{TDC} = 1.02$ – 1.39 MPa; (blue) $p_0 = 80.0$ kPa, $p_{TDC} = 1.23$ – 1.57 MPa. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

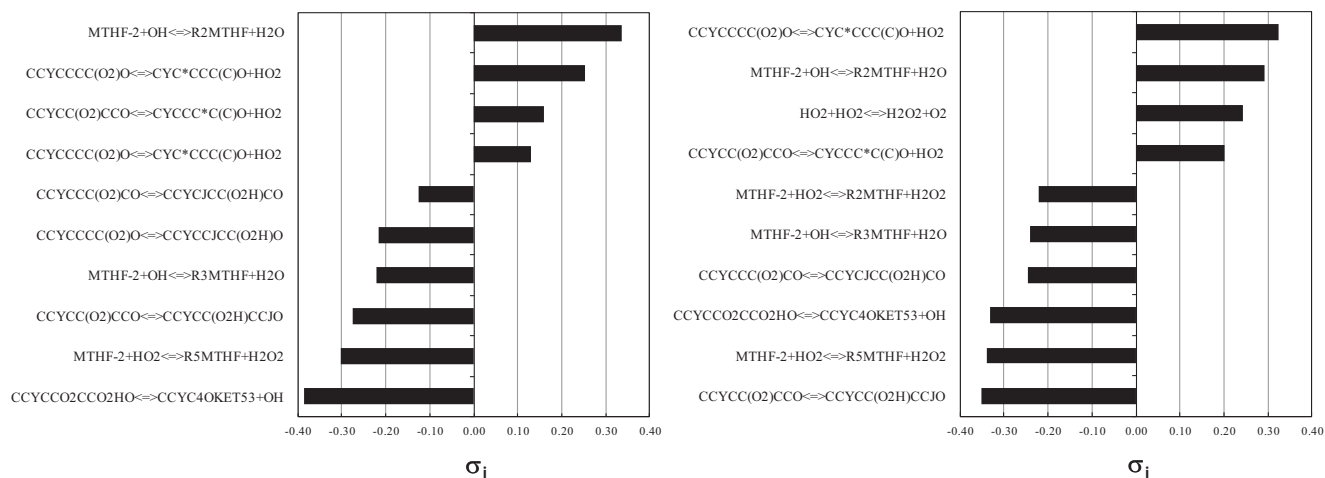


Fig. 6. Brute force sensitivity analysis of 2-MTHF/air mixture on first stage ignition delay (left side) and on the total ignition delay (right side). $\phi = 1.0$, $p_{TDC} = 0.67$ MPa, $T = 719$ K. Species names are described in Table 2.

4.2. Reaction intermediates

To provide a better understanding of the kinetics of the oxidation of 2-MTHF in the low-temperature range, the reacting mixture was extracted from the reaction chamber into a previously vacuumed heated sampling canister at selected times after TDC. Each sampling experiment therefore corresponds to a separate compression event. The volumetric ratio between the combustion chamber and the canister is superior to 40, and the following fast decrease in pressure and temperature ensures instantaneous quenching of the reactivity. A stoichiometric 2-MTHF/O₂/N₂ mixture was used, at $p_{TDC} = 0.76$ MPa and $T_c = 719$ K. In these conditions, the first stage and total ignition delays are 28.7 ms and 40.0 ms, respectively.

Identification and quantification of the stable intermediate species was performed from the acquired samples. Identified compounds are listed in Table 6 and a representative chromatogram is

presented in Fig. 7. H₂, CO, CO₂, formaldehyde and water were not measured.

The largest peaks on the chromatogram are attributed to unsaturated counterparts of 2-MTHF as methyldihydrofurans, 1-cyclopropyl-ethanone, pentenone or 2-methylfuran (products 17–20) and usual products of the combustion of middle-sized oxygenated compounds (products 1–16). With increasing retention times, specific low temperature intermediates, resulting from the addition of 2-MTHFyl radicals on O₂, followed by intramolecular H-transfer and loss of $\cdot$ OH appear: 2-methyltetrahydrofuranone isomers, 1-propen-2-ol acetate and allylacetate.

The analysis of the initial mixture was compared with results at top dead center to ensure the absence of reactivity and fuel condensation during the compression phase. The absence of significant quantities of intermediates at TDC, as visible in Fig. 8, also provides proof of the absence of reactivity during the compression phase. A comparison of experimental and modeled mole fraction

Table 6

Stable intermediate species identified during the ignition delay of a 2-MTHF/O₂/N₂ mixture. $T_C = 719$ K and $p_{TDC} = 0.76$ MPa.

No. in Fig. 7	Species	Molar mass	Structure
1	C ₁ –C ₂ species	–	CH ₄ , C ₂ H ₆ , C ₂ H ₄ , C ₂ H ₂
2	Propene	42	
3	Propane	44	
4	C ₃ H ₄ isomers	40	
5	Acetaldehyde	44	
6	1-butene	56	
7	1,3-butadiene	54	
8	2-butene isomers	56	
9	Methyloxirane	58	
10	2-Propenal	56	
11	Furan	68	
12	Propanal	58	
13	Acetone	58	
14	(2,3)-dihydrofuran	70	
15	(2,3)-epoxybutane	70	
16	Methylvinylketone	68	
17	2-methylfuran	82	
18	2-Butanone	72	
19	3-pentenal	84	
20	1-cyclopropylethanone	84	
21	(2,3)-dihydro-5-methylfuran	84	
22	2-methyltetrahydrofuran	86	
23	Allylacetate	100	
24	Unidentified		
25	2-methyltetrahydrofuranone isomers	100	
26	1-propen-2-ol acetate	100	

profiles for the quantified stable intermediates is plotted in Fig. 8. In these conditions, the simulated ignition delay is 47.8 ms while the experimental ignition delay is 40 ms. The measured first stage ignition delay is 22.7 ms and the calculated one of 23.8 ms. Note that given the small difference between simulated and experimental delays, the profiles were not normalized timewise.

As depicted in Fig. 8, the predicted fuel profile respects the trend and values of measured mole fractions, except for the decay

rate of the fuel mole fraction, which is slightly underpredicted by the model at ignition. The trend of simulated profiles of the most abundant of the unsaturated counterparts of 2-MTHF is in fair agreement with the experimental data, with a maximal predicted mole fraction overestimated by a factor 4 for (2,3)-dihydro-5-methylfuran. However, as shown on the chromatogram (Fig. 7), the peak corresponding to (2,3)-dihydro-5-methylfuran (#21 in Fig. 7) is not fully separated from the broad peak of the fuel, leading

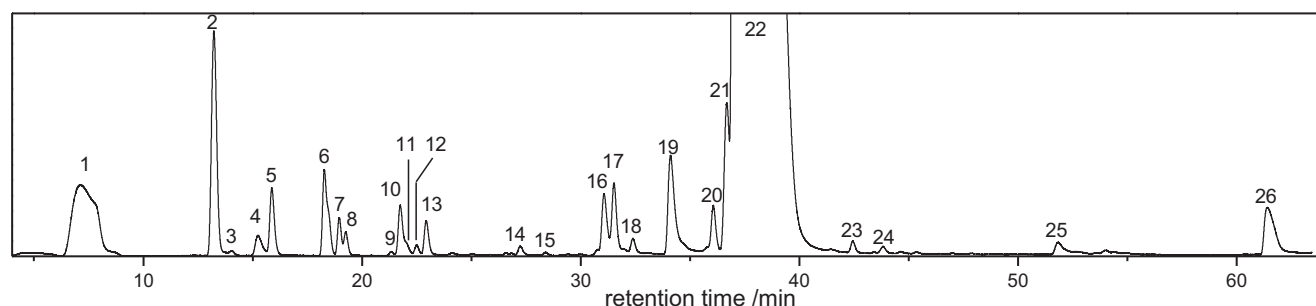


Fig. 7. Representative chromatogram of the stable intermediate species formed during the two-stage ignition of a stoichiometric 2-MTHF/O₂/N₂ mixture, with a sampling time = 34 ms. $T_c = 719$ K and $p_{TDC} = 0.76$ MPa.

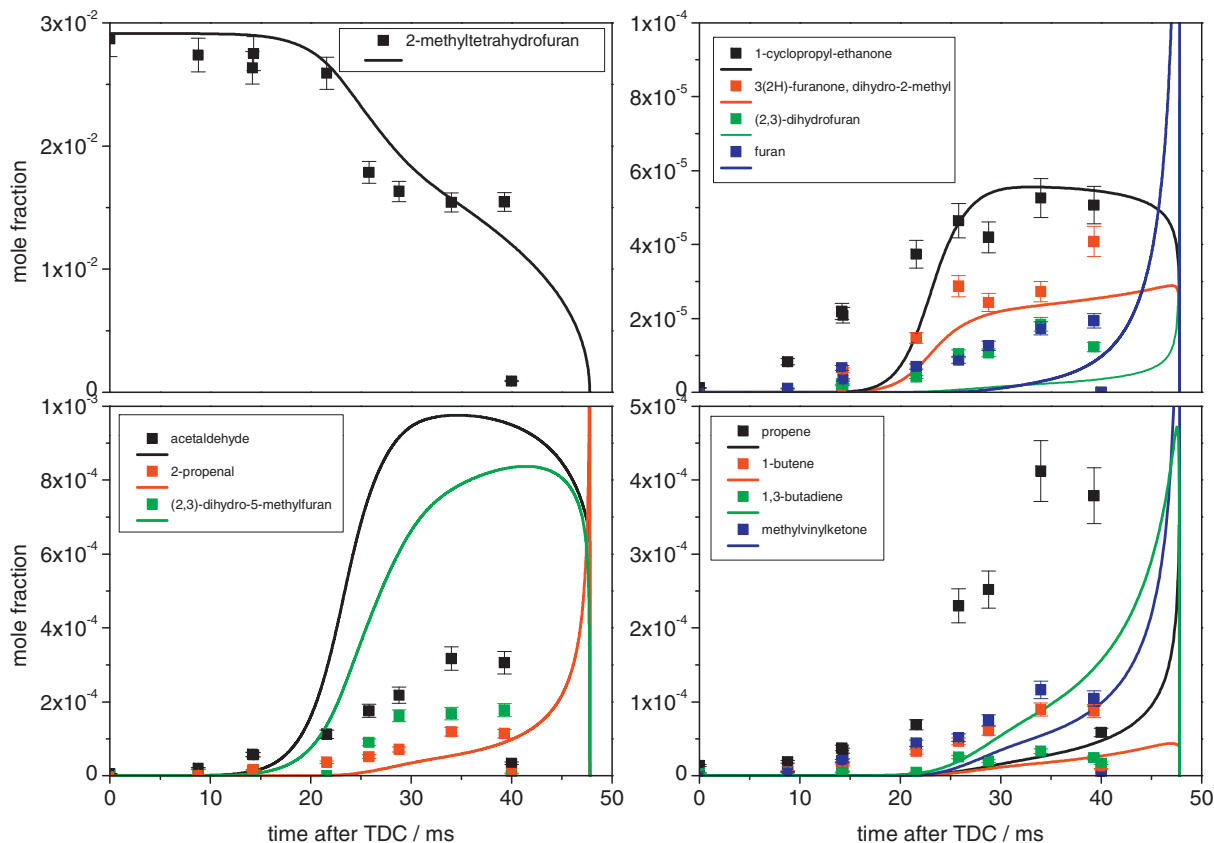


Fig. 8. Mole fraction - time profiles of 2-MTHF and stable oxidation intermediate species for a stoichiometric 2-MTHF/O₂/N₂ mixture, $T_c = 719$ K, $p_{TDC} = 0.76$ MPa.

to a possible underestimation of its experimental mole fraction. 1-Cyclopropyl-ethanone (#20 in Fig. 7) formed directly from (2,3)-dihydro-5-methylfuran is well represented by the model, as well as dihydro-2-methyl-3-furanone (#25 in Fig. 7). These observations prove the overall reliability of the description of the primary reaction pathways of the oxidation of 2-MTHF by the kinetic model.

Other simulated intermediates species show globally profiles that respect the observed trends with, yet, only qualitative agreement. It is the case of unsaturated species 1-butene (#6 in Fig. 7), propene (#2 in Fig. 7), 1,3-butadiene (#25 in Fig. 7) and methylvinylketone (#16 in Fig. 7). Discrepancies in the absolute mole fractions of small species could be due to missing ketohydroperoxide and bicyclic ethers decomposition pathways. Further work is needed to investigate such reactions. However, these potential missing steps do not deteriorate the ability of the model to simulate the reactivity of 2-MTHF, as these pathways are not rate limiting.

4.3. Low temperature reaction pathways

In order to provide a better understanding of the preferential reaction pathways of 2-MTHF in the low temperature range, the main reaction paths of the oxidation of 2-MTHF were deduced from the simulation results and are depicted in Fig. 9. This was computed as a RCM experiment, at a time of 25.8 ms after the top dead center time, corresponding to 2 ms after the first stage ignition time, for $p_{TDC} = 0.76$ MPa and $T_c = 719$ K corresponding to the sampling experiments. At the chosen time the calculated pressure is 0.79 MPa and the temperature is 757 K. At this point, about 25% of the fuel is converted. In Fig. 9, the first steps of the oxidation of 2-MTHF in these conditions are reported. Percentages represent the proportion of the species reacting through the considered reaction.

The scheme is divided into 5 regions, for the 5 types of MTHFyl' radicals directly formed after H-abstraction from 2-MTHF.

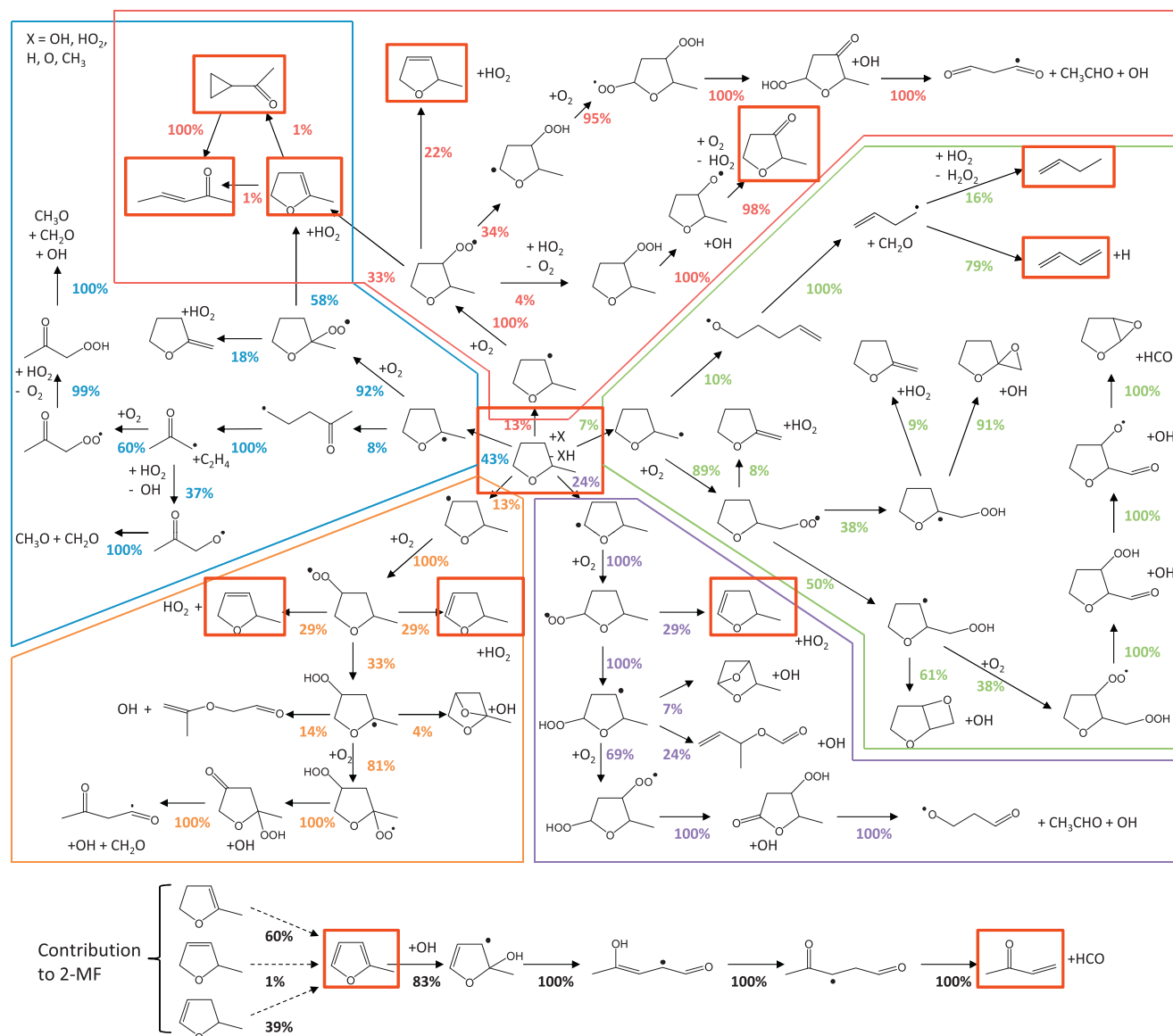


Fig. 9. Main reaction paths of the oxidation of 2-MTHF in a RCM at a time of 25.8 ms after TDC, with $T = 757$ K and $p = 0.76$ MPa.

The distribution of these radicals is correlated with the BDE of the C–H bonds presented in Fig. 1, as well as the number of available hydrogen atoms. Logically, the R2MTHF radical is formed at a higher rate than others, with more than 40% of the rate of consumption of 2-MTHF. Then, R5MTHF represents 24% of the consumption of 2-MTHF. R3MTHF and R4MTHF are produced in the same amounts, 13%. The formation of CH2MTHF can be considered a marginal reaction pathway with 7% of rate of production. Following the simulation results in these conditions, small amounts of CH2MTHF and R2MTHF can react by ring opening, leading, for CH2MTHF, to the production of 1-butene and 1,3-butadiene, and for R2MTHF, to the pentan-2-on-5-yl radical. Globally, MTHFyl' radical addition on O_2 is the main reaction pathway. From the formed MTHFylOO' radicals, high amounts of methylenetetrahydrofuran and methylidihydrofurans are obtained, the one being formed in the largest amount being 5-methyl-(2,3)-dihydrofuran. 5-methyl-(2,3)-dihydrofuran can isomerize into 1-cyclopropylethanone or pent-3-en-2-one [29]. Unsaturated cyclic ethers can yield 2-methylfuran. Decomposition of 2-methylfuran leads mainly to the formation of 3-buten-2-one. MTHFylOO' radicals can also isomerize into *MTHFOOH, form bicyclic ethers, methyltetrahydro-

furanones or, after a ring opening step, yield unsaturated esters, as described in Fig. 9. Following reactions are second O_2 additions, formation of ketohydroperoxides and finally decomposition into smaller species. The species that were observed experimentally are circled in red in the scheme.

4.4. High temperature model validation

In order to extend the validation of the model to the high temperature regime, it was used to simulate experimental data found in the literature.

Moshammer et al. [7] studied low pressure 2-MTHF/ O_2 /Ar burner-stabilized flat laminar flames at an equivalence ratio of 1.7. Concentration profiles of reactants, diluent and intermediate species were measured with two independent MBMS (Molecular Beam Mass Spectrometry) instruments: Electron Impact-MBMS in Bielefeld and photoionization-MBMS at the Advanced Light Source in Berkeley. The authors used their experimental data to validate a kinetic model of the oxidation of 2-MTHF in the high temperature regime. Experimental mole fraction profiles of some species quantified with EI-MBMS are presented as a function of

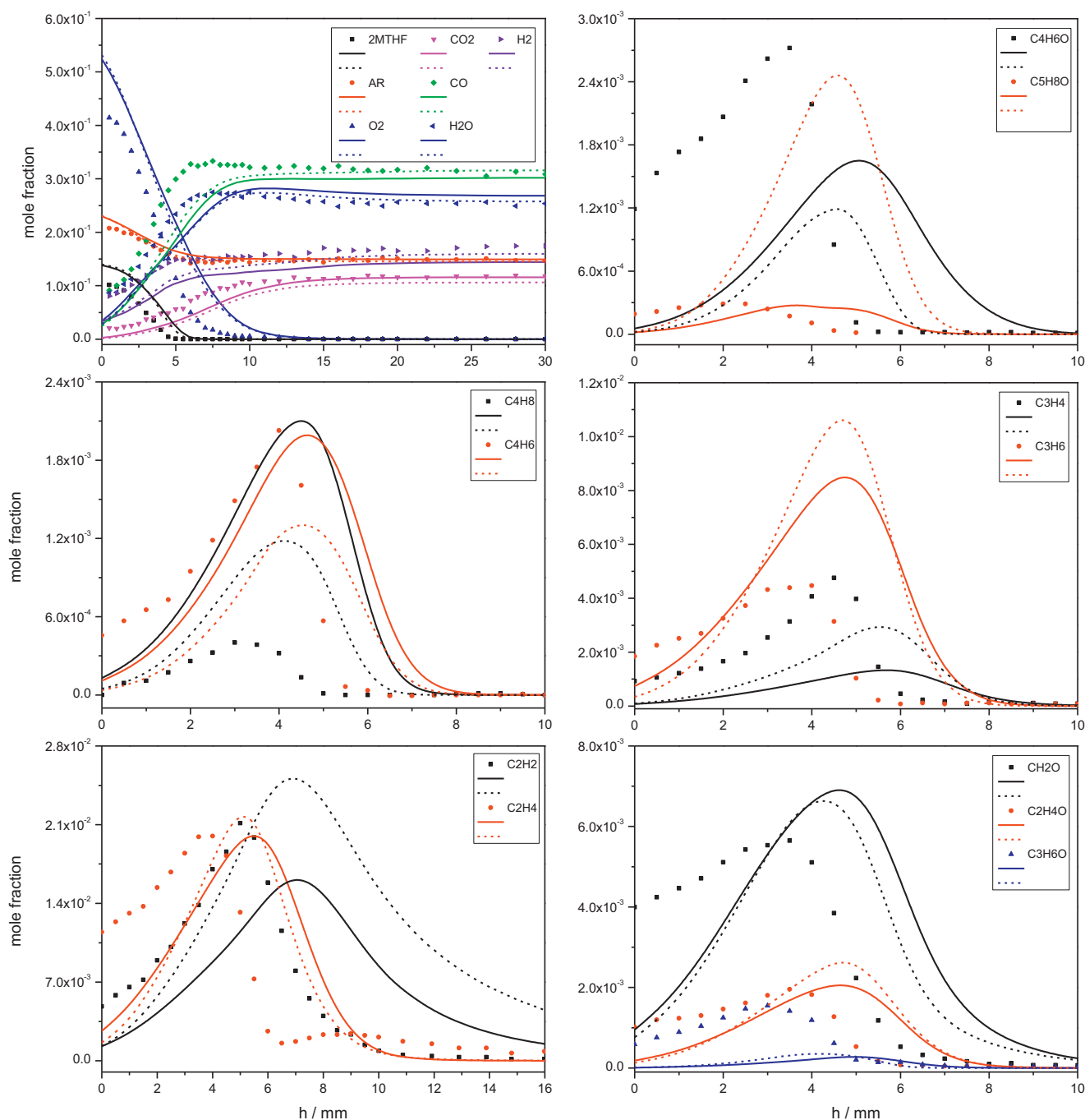


Fig. 10. Mole fraction profiles for reactants and stable intermediate species in a 2-MTHF (0.33 standard liter per minute)/O₂(1.3767 slm)/Ar(0.5689 slm) fuel-rich flame, at $P = 4$ kPa, experimentally measured by EI-MBMS (symbols) [7] as a function of the distance to the burner. Simulation results from the model from Moshammer et al. [7] (dashed line) and the current model (continuous line).

the distance from the burner in Fig. 10, and compared with simulation results from the original and current models. The measured 2-MTHF mole fraction directly at the exit of the burner is lower than the initial mole fraction of the mixture, while intermediate species, as formaldehyde, are present in significant amounts. These observations suggest the possibility of reactivity inside the burner or probe-induced perturbations (600 μ m diameter orifice). Uncertainties due to probe effects in low pressure flame samples have been discussed by Gururajan et al. [37]. Helped by direct numerical simulations taking into account the effects of the probe and supporting flange, the sampling position and compressibility

of the gases, Gururajan et al. concluded that the mole fraction profiles could be distorted.

Globally, the current model shows comparable performance in the prediction of the experimental data to the model from Moshammer et al. [7]. This is visible for reactants (2-MTHF and O₂) as well as for combustion products (CO, CO₂ and H₂O) mole fraction profiles. 2-MTHF is converted for short distances from the burner, even directly at the exit of the burner, and the mole fractions of final combustion products reach a plateau of concentration above 8 mm. Predicted mole fraction profiles of abundant intermediates such as 1,3-butadiene, ethylene, acetalde-

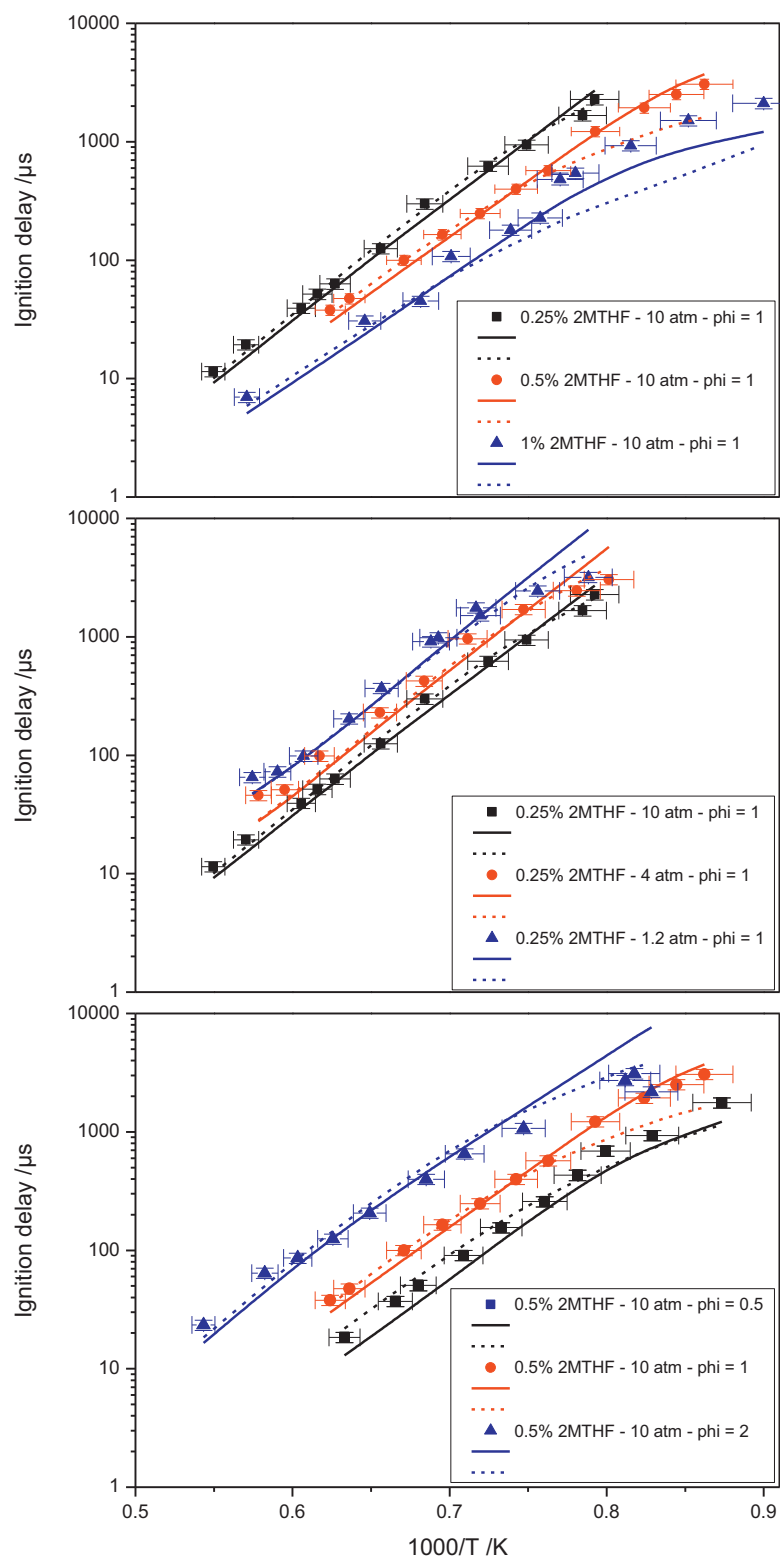


Fig. 11. Experimental (symbols) and simulated (Moshammer et al.: dashed line; current model: continuous line) shock tube ignition delays as a function of the temperature for different fuel dilutions, pressures and equivalence ratios, as reported in [8].

hyde, formaldehyde and C_5H_8O isomers are in a good agreement with the experimental data. Acetylene and propene maximum mole fraction are fairly well reproduced. Nevertheless, the mole fractions of C_3H_4 and C_3H_6O isomers are underestimated. The main improvement of the current model lies in a severe reduction of the concentration maxima of C_5H_8O isomers, which now match

the current experimental data. Both models predict lower mole fractions for C_3H_6O isomers (mainly acetone and 2-propanal) than observed experimentally, suggesting missing high temperature formation pathways leading to these species. Globally, changes made to the C_0 – C_4 subsection of the model did not deteriorate the performance of the model in the high temperature regime. Changing

the C₀–C₄ subsection can affect the distribution of small species, but the most sensitive modification in the high temperature mechanism was the correction of the reaction rates of 2-MTHF + •OH yielding 2-methyltetrahydrofuranyls, as discussed in Section 3. This led to different concentrations of small species compared to the ones calculated by Moshhammer et al. [7]. Also, the formation of 2-methylfuran (2-MF) was not included in Moshhammer et al.'s model and its formation was not reported in the studied flames. The maximal mole fraction of 2-MF, in a flat flame as described by Moshhammer et al. [7], predicted by the current model is 3.5×10^{-4} , the same order of magnitude than the experimentally observed mole fractions of butene isomers or C₅H₈O isomers.

Wang et al. [8] measured the ignition delays of 2-MTHF/O₂/Ar mixtures behind reflected shock waves. The influence of the temperature (1000–1900 K), equivalence ratio (0.5–2), pressure (1.2–10 atm) and fuel dilution (0.25–1%) were investigated. The authors reported good agreement between their experimental data and simulations from the model developed by Moshhammer et al. [7]. Both experimental ignition delays and modeling results from both models are plotted in Fig. 11. The uncertainties of temperature and ignition delay in these experiments are estimated by the authors to be ± 25 K and $\pm 10\%$ individually, and are reported in Fig. 11.

Overall, the model is in good agreement with the ignition delays obtained for the three investigated pressures, 1.2, 4 and 10 atm, and equivalence ratios from 0.5 to 2 match with experimental data. Especially, the agreement is superior to Moshhammer et al.'s model [7] at the lowest investigated temperatures.

5. Conclusions

Low temperature combustion of 2-MTHF/O₂/inert stoichiometric mixtures was investigated at high pressures and temperatures below 900 K. To do so, new experimental ignition delays, along with first stage ignition delays when relevant, were measured for top dead center pressure ranging from 0.3 to 2.1 MPa and core gas temperatures between 640 and 900 K, in the U. Lille Rapid Compression Machine. A deviation from Arrhenius behavior of the ignition delays with temperature was observed between 750 and 850 K. First stage ignition delays were recorded at temperatures from 685 to 790 K. In order to investigate the low temperature behavior of 2-MTHF, samples of the reacting mixture were extracted at different times between top dead center and ignition, enabling to measure the mole fraction profiles of the stable intermediate species during two stage ignition at $T_c = 719$ K and $p_{TDC} = 0.76$ MPa.

A detailed kinetic model describing both low and high temperature regimes of the oxidation of 2-MTHF was built. It consists in 507 gas phase species involved in 2425 reactions. The model was validated by comparison of the simulations with new experimental data. It is able to predict first stage and total ignition delays, as well as mole fraction profiles of the fuel and most primary stable intermediates species that were quantified in the experiments with reasonable agreement. A reaction pathway analysis and brute-force sensitivity analysis show that the prediction of the first stage ignition delay is strongly correlated with the balance between the different radicals formed through H abstraction on the fuel, the tertiary radical R2MTHF being unable to lead to low temperature branching. On the other hand, the formation of R5MTHF strongly favors first stage ignition as the RO₂• adduct formed from this radical can react through numerous internal isomerization pathways which further lead to low temperature branching. In contrast, the formation of R2MTHF from the 2-MTHF + HO₂• pathway facilitates ignition, as it constitutes an important source of H₂O₂.

The validation of the kinetic model was extended to the high temperature regime by comparing the simulation results with mole fraction profiles from burner stabilized laminar flat flames

and high temperature ignition delays in a shock tube, showing good agreement with the experimental data.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2017.01.008.

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